

Guavion: An Application Agnostic AI-Orchestrated Platform for End-to-End Workflows

Executive Summary

Guavion is an AI-orchestrated, web/app-based platform that turns complex, multi-step work into repeatable end-to-end workflows. It is built for teams whose outcomes depend on coordinating data, documents, tools, approvals, compute, and analysis across many systems and stakeholders. Guavion replaces the brittle “glue layer” of scripts, spreadsheets, dashboards, copy-paste procedures, and one-off automations with a shared workspace where users state intent in plain language, review a proposed plan, execute it safely through audited tools, and receive structured outputs with complete provenance.

Molecular simulation is one flagship use case because it clearly demonstrates the platform’s core challenges and value: translating goals into validated protocols, preparing complex objects (molecules and materials), orchestrating execution across local/HPC/cloud, and producing decision-ready analyses from many runs. Guavion already demonstrates key platform primitives through a live demo at <https://chopchopmol.com/>, including fast web-based visualization for very large systems (for example ~500K atoms), database retrieval (for example PubChem and PDB), project storage, ML potential integration (for example MACE), and an AI agent that can perform multi-step tasks (for example torsion scans) and analyze results. A short introduction video is available at https://www.youtube.com/watch?v=YaDyhtO_sAQ.

However, Guavion is not a “chemistry product that expands later.” Chemistry is a proof point and an initial vertical. The core product is the Guavion platform: a general workflow operating layer that can support domain apps for banking and finance, real estate, engineering, and other technical and operational functions. Each domain is implemented as validated templates, tool connectors, and output schemas on top of the same orchestration, provenance, and collaboration backbone.

Vision

Provide every technical team with a real-time, interactive AI copilot that plans, runs, and explains end-to-end workflows directly inside a shared visual workspace, with reliable execution, strong guardrails, and auditable provenance.

The Problem: High-Value Work Is Still Procedural

Across industries, high-impact work remains procedural and tool-fragmented. Teams translate goals into a sequence of steps spanning multiple software products, internal systems, scripts, approvals, and compute environments. This produces universal pain points:

1) Fragmented Tooling and “Workflow Glue.”

Teams stitch together spreadsheets, notebooks, internal portals, third-party SaaS tools, scripts, and specialized GUIs. The workflow exists as tacit knowledge distributed across people rather than as a reusable, validated process. Changes in personnel, vendor tools, or infrastructure break the workflow.

2) Slow and Error-Prone Translation of Intent into Execution

Even when a team knows what it wants to do, the cost is in “how”: how to format inputs, where to run it, how to parameterize it, and how to ensure results are correct and comparable. Human coordination and manual steps cause delays and silent errors.

3) Poor Standardization and Reuse

Organizations rarely have a unified library of approved workflows. Instead, they have variations of scripts and ad hoc practices. This prevents scaling best practices, onboarding quickly, or enforcing consistency.

4) Weak Traceability and Auditability

Decisions often rely on assumptions, parameter choices, and intermediate transformations that are not fully recorded. When something changes later inputs, market conditions, and simulation settings, teams cannot reproduce results confidently or explain why outcomes differ.

5) Analysis and Reporting as a Tax

The final stage is often the most time-consuming: collecting outputs, producing comparisons, making plots, generating reports, and explaining results. This is repeated constantly, often from scratch.

What’s missing is a guided layer that turns intent into validated workflows, executes them across the right tools and compute, and produces decision-ready results with complete provenance.

The Guavion Platform: Core Components

Guavion combines a shared workbench, an AI agent constrained by audited tool-calling, and an orchestration + provenance engine.

1) Shared Web-Based Workbench

Guavion provides a single browser-based workspace for work that typically spans many systems.

- **Projects and artifacts:** Every workflow lives within a project that contains structured artifacts (datasets, documents, objects like molecules or designs, workflows, runs, results).
- **Collaboration:** Shared access, comments, and controlled sharing of results and intermediate steps.
- **Visual interaction:** A visual representation of objects (molecules, portfolios, properties, designs) and workflow graphs (steps, dependencies, outputs).
- **Anywhere access:** Cloud-backed workspaces with optional local execution integration.

2) The AI Agent: Operator + Advisor

Guavion's agent is not just chat. It is an operator that can execute work using a controlled set of platform tools, and an advisor that explains tradeoffs and risks.

- **Planning:** Converts natural-language goals into a workflow plan with steps, dependencies, and outputs.
- **Validation:** Checks for missing information, incompatible configurations, and risky assumptions.
- **Cost preview:** Estimates the runtime and compute cost (or operational cost) before execution.
- **Execution:** Runs the workflow by calling audited tools (connectors, engines, and platform actions).
- **Failure recovery:** Diagnoses common failure modes, proposes fixes, and logs all changes.
- **Explanation:** Produces clear reasoning and summaries for users, stakeholders, and audits.

The critical design choice: the agent is constrained by audited functions and templates, which makes execution reliable, governable, and repeatable.

3) Workflow Templates and Guardrails

Guavion standardizes best practices as templates.

- **Templates encode “how we do this”** for a given task with validated defaults and required inputs.
- **Guardrails enforce safety** (policy checks, validation rules, cost ceilings, and approval steps).
- **Versioning enables change control:** updates to workflows are tracked and can be rolled back.

This is the mechanism that turns tribal knowledge into institutional capability.

4) Orchestration Across Systems and Compute

Guavion runs workflows across heterogeneous environments while keeping execution semantics consistent.

- **Local execution** for lightweight tasks and prototyping.
- **HPC schedulers** for heavy compute, with standardized job templates and monitoring.
- **Cloud compute** for burst capacity, GPUs, and scalable batch workflows.
- **Enterprise systems and APIs** via connectors (data warehouses, CRMs, document stores, internal tools).
- **Observability:** logs, metrics, and step-level status for every run.

5) Provenance Graph, Search, and Decision-Ready Outputs

Guavion’s provenance layer records the full story of work.

- **Inputs, transforms, parameters, tool versions, compute context, and outputs** are all linked.
- **Search and reuse:** retrieve prior runs, clone workflows, compare differences.
- **Batch analytics:** aggregate across runs and produce standardized comparisons.
- **Reporting:** automatically generate narrative summaries and stakeholder-ready deliverables with links back to source provenance.

How Guavion Works in Practice

A typical Guavion workflow looks like this:

1. **User intent:** “Do X, using Y constraints, and compare results to Z.”
2. **Agent plan:** Guavion proposes a workflow plan (steps, tools, resources, expected outputs).

3. **Validation + cost preview:** Guavion checks missing inputs, policy constraints, cost bounds, and risks.
4. **Approval:** User or organization approves execution (optionally step-gated).
5. **Orchestration:** Guavion dispatches tasks to the appropriate backends (local/HPC/cloud/systems).
6. **Monitoring:** Real-time status with failure detection and suggested fixes.
7. **Structured output:** Results captured as tables, visualizations, and reports.
8. **Provenance:** Everything is logged, searchable, and reusable.

Detailed Use Cases Across Domains

Below are detailed user journeys showing how the same Guavion platform supports different industries via templates + connectors.

Use Case A: Chemistry and Materials (Flagship Vertical)

Who uses it

Computational chemists, materials modelers, drug discovery computational groups, and research teams that run repeated simulation protocols across many compounds/systems.

The workflow problem today

- Complex setup: structure prep, charge/spin, constraints, method selection
- Engine fragmentation: multiple QC/MD codes, each with different inputs
- Compute fragmentation: local prototyping, HPC production, cloud GPU for ML potentials
- High failure rates: convergence issues, bad geometries, incorrect settings
- Heavy post-processing: parse logs, compute metrics, compare across series, generate plots

What Guavion does end-to-end

1. **Ingest and prepare structures**
 - Retrieve from PubChem/PDB or upload files
 - Standardize and validate: missing atoms, charges, stereochemistry, periodicity
 - Generate conformers, enumerate tautomers/protonation states if needed
 - Record all transformations in provenance

2. **Select a validated protocol**

User: "Geometry optimize, then compute energies and dipoles for this series."

Guavion: proposes method options (accuracy vs cost), convergence thresholds, dispersion/solvent settings, with expected runtime and compute requirements.

Guardrails: flags risky configurations (e.g., ill-conditioned SCF, unrealistic constraints).

3. **Execute across compute**

- Lightweight stages local, heavy stages HPC, GPU stages cloud
- Standard job templates, monitoring, and structured logs
- Automatic retries and failure triage suggestions

4. **Aggregate results and compare**

- Parse outputs into structured tables: energies, gaps, dipoles, convergence metrics
- Create plots across the series and compare against historical baselines
- Produce a decision summary: trends, outliers, uncertainty signals, rerun suggestions

5. **Reusable templates for teams**

Teams can save "approved workflows" so new members can run standard protocols safely with one click, while maintaining audit trails.

Use Case B: Banking and Finance

Who uses it

Risk teams, quant teams, portfolio managers, credit underwriting, capital markets analytics, and financial ops groups.

The workflow problem today

- Data lives across warehouses, spreadsheets, vendor feeds, and internal systems
- Processes are gated by approvals and policy constraints
- Models must be consistent and auditable
- Reporting is repetitive and time-sensitive

What Guavion does end-to-end

1. **Data retrieval and normalization**

User: "Run a stress test on these portfolios with three macro scenarios."

Guavion connects to approved data sources, pulls holdings and risk factors, logs data snapshots, and validates completeness.

2. **Execute validated templates**

Guavion runs a governed template (scenario construction, shocks, model selection, calculation steps), enforcing policy rules like approved models, parameter bounds, and sign-off steps.

3. **Attribution and explanation**

Guavion produces:

- Loss distributions, VaR/ES changes, scenario P&L
- Attribution to drivers (rates, spreads, FX, sector exposures)
- Narrative explanation suitable for stakeholders

4. **Audit trail and reproducibility**

Every run stores: data version, scenario parameters, model versions, who approved, and output artifacts. A future reviewer can reproduce the result exactly.

5. **Deliverables**

Guavion outputs a board-ready report and a structured dataset for downstream systems, with links to provenance.

Use Case C: Real Estate

Who uses it

Acquisitions teams, asset management, underwriting analysts, lenders, and investment committees.

The workflow problem today

- Data rooms contain messy documents (leases, rent rolls, T12s, appraisals)
- Underwriting requires assumptions and sensitivity analysis
- Teams collaborate across multiple tools and spreadsheets
- Due diligence is repetitive across deals

What Guavion does end-to-end

1. **Ingest documents and extract key fields**

User: “Underwrite these five properties and flag lease risks.”

Guavion ingests the data room, extracts structured fields (rent roll terms, escalations, expirations), and highlights anomalies.

2. **Run underwriting templates**

Guavion executes a standardized underwriting workflow: NOI normalization, cap rate assumptions, debt sizing, DSCR, and sensitivity cases.

3. **Scenario analysis**

Run rent growth, vacancy, expense inflation scenarios, and compare outputs side-by-side.

4. **Risk flags and explanations**

Guavion outputs:

- Key risk factors (tenant concentration, near-term expirations, unusual clauses)
- Sensitivity “breakpoints” (what assumptions break the deal)
- A comparable deal summary pack

5. **Provenance for assumptions**

Every assumption is logged, versioned, and attributable. Teams can reuse prior deal templates and adapt quickly.

Use Case D: Engineering and Simulation

Who uses it

Mechanical, aerospace, civil, and product engineering teams running simulations, optimization studies, and design-of-experiments.

The workflow problem today

- Parameter sweeps and simulations require manual scripting
- Compute environments are fragmented
- Result aggregation is tedious
- Teams struggle to standardize best practices

What Guavion does end-to-end

1. **Define the design space**

User: “Sweep these parameters and rank designs under cost and safety constraints.”

Guavion builds a workflow plan with parameter ranges, constraints, and outputs.

2. **Orchestrate simulation batches**

Dispatch large simulation sets to HPC/cloud with standardized job templates, monitoring, and automatic resubmission rules.

3. **Aggregate results and rank tradeoffs**

Guavion produces Pareto fronts, constraint violation reports, and ranked recommendations. Results are linked to the exact simulation settings and code versions.

4. Reusable playbooks

Organizations maintain validated templates for common studies, speeding onboarding and reducing error rates.

Why Guavion Is Defensible

Guavion's moat is not "chat." It is the combination of:

- **Workflow templates that encode best practices**
- **Audited tool-calling execution** rather than ad hoc scripts
- **Orchestration across environments** with consistent semantics
- **Provenance graph and institutional memory** that grows with use
- **Reusable reporting and comparison tooling** tied to history and governance

The more a team uses Guavion, the more valuable its template library, execution telemetry, and searchable workflow history become.

Initial Users and Wedge

Guavion launches with a flagship vertical in chemistry/materials because it is a high-friction domain that strongly benefits from orchestration, provenance, and analysis at scale. Early design partners validate reliability, guardrails, and "time-to-insight" improvements.

In parallel, Guavion validates one additional vertical (finance, real estate, or engineering) using the same core platform to prove cross-domain generality. Expansion is achieved by adding template libraries and connectors, not by rebuilding the platform.

Pre-Seed Plan: Team and Goals

We seek pre-seed funding to take Guavion from demo to v1 and to establish early commercial traction.

Key roles

- **Backend / Orchestration Lead:** workflow engine, APIs, data model, provenance, connectors
- **Full-Stack / Product Engineer:** workbench UX, visualization, collaboration, workflow builder
- **Infra / Security Engineer:** deployments, monitoring, RBAC/SSO, scheduler + cloud integrations

- **Flagship Vertical Lead (Chemistry):** validated protocols, benchmarks, design partner success
- **Second Vertical Lead (Finance/RE/Engineering):** template design, connector requirements, pilot success

Goals for this round

- Production-quality v1 platform core (workbench + orchestration + provenance)
- Flagship chemistry workflows shipped end-to-end with guardrails and batch analysis
- At least one additional vertical template library piloted with design partners
- Early paid pilots demonstrating repeatable value

Product Milestones

Phase 1: v1 Platform + Flagship Vertical (Months 0–6)

- Harden projects/artifacts/workflow model
- Ship a small set of validated chemistry templates (for example geometry opt, single-point, torsion scans, batch comparisons)
- Connect to real HPC/cloud execution with monitoring
- Constrained AI assistant with validation, cost previews, and safe execution

Phase 2: Multi-Backend + Multi-Vertical (Months 6–12)

- Expand engine/backends and reliability features
- Team workflow features: shared dashboards, approvals, search and reuse
- Pilot second vertical templates with 3–5 design partners
- Improve reporting and comparison across runs and projects

Phase 3: Early Commercial Traction (Months 12–24)

- Enterprise hardening: RBAC, SSO, audit logs, security posture
- Scalable onboarding and repeatable deployment
- Convert pilots to paying customers across multiple domains
- Demonstrate traction and expansion to support a Seed round